MEMORANDUM

TO: Professor Kid Kidd

FROM: Krutika Patel

DATE: 10/31/2026

SUBJECT: Arduino Voltage Meter

1. Objective

The objective of this laboratory experiment was to develop a voltage device that measures input voltage and displays the measured value on an LCD. This experiment also involved reviewing the LCD and ATmega328P microcontroller specifications to determine their operating limits, capabilities, and compatibility.

2. Components Used

The components used in this experiment included:

- Arduino Uno (ATmega328P microcontroller)
- 16×2 LCD Display with I2C Interface
- Breadboard
- Resistors
- Jumper Wires
- USB Cable and Computer

3. Experimental Design and Discussion

3.1 Do we need an External Power Source?

An external power supply is not required because the Arduino receives power from the computer through a USB cable. The Arduino also provides 5 V to power the LCD; however, measuring a separate voltage source requires that both the circuit and the Arduino share a common ground.

3.2 What libraries do we need?

Two Arduino libraries are used to communicate with the LCD: Wire.h and LiquidCrystal_I2C.h.

3.3 What communication protocol are we using and why?

The communication protocol used in this experiment is Inter-Integrated Circuit (I2C). It allows the Arduino to communicate with the LCD using only two pins: Serial Data (SDA) and Serial Clock (SCL). I2C was chosen because it reduces the number of wires needed, makes the circuit easier to connect, and leaves other pins available for voltage measurements.

3.4 Which pins are connected to get output?

The LCD connects to the Arduino Uno through four main connections:

LCD Pin	Arduino Connection	Purpose
GND	GND	Common electrical ground
VCC	5V	Supplies power to the LCD
SDA	A4	Serial data communication
SCL	A5	Serial clock communication

In addition to these connections, analog input A0 is used to receive the voltage signal from the circuit. The input converts the voltage at A0 into a digital value that can be processed and displayed on the LCD.

3.5 Circuit Design and Explanation

The circuit consists of an Arduino Uno, an I2C LCD, and a resistor network connected through a breadboard. The LCD connects to the Arduino via the I2C interface, while the circuit connects to analog input A0.

For measurements involving an input voltage higher than the ADC reference voltage, a voltage divider can be used to reduce the voltage to a safe measurement range.

The voltage divider relationship is:

$$V_{out} = V_{in} \times R2 / (R1 + R2)$$

For a standard 5 V Arduino Uno configuration, the voltage at A0 must remain within the permitted analog input range. Assuming the reference voltage, the measured voltage at the analog input can be calculated using:

$$V_{out} = (\text{ADC Reading} / 1023) \times 5.0\ V$$

This design allows the Arduino to process an analog voltage and present the corresponding measurement on the LCD.

4. Experimental Procedure

1. Connect the Arduino Uno to the computer through a USB cable.

2. Connect the I2C LCD module to the Arduino.
3. Install the required LiquidCrystal_I2C library through the Arduino IDE.
4. Upload the Hello World program provided to verify that the LCD and I2C communication are functioning.
5. Assemble the resistor measurement circuit on the breadboard.
6. Connect the measurement circuit output to analog input A0.
7. Upload the voltage measurement program to the Arduino.
8. Observe the displayed voltage and compare it with the expected input voltage.

5. Results and Observations

The LCD backlight illuminated when the Arduino was connected to the computer through USB. Characters were also visible on the LCD, indicating that the display was receiving power and producing an output.

The final measured voltage and its accuracy must be determined from the completed experiment.

Measured Voltage(s): 2.49 V and 1.66 V

Expected Voltage(s): 2.49 V and 1.66 V

The following figures show the circuit setups for Case 1 and Case 2. The two cases demonstrate different circuit configurations used for voltage measurement.

Figure 1: Arduino Voltage Measurement Circuit – Case 1

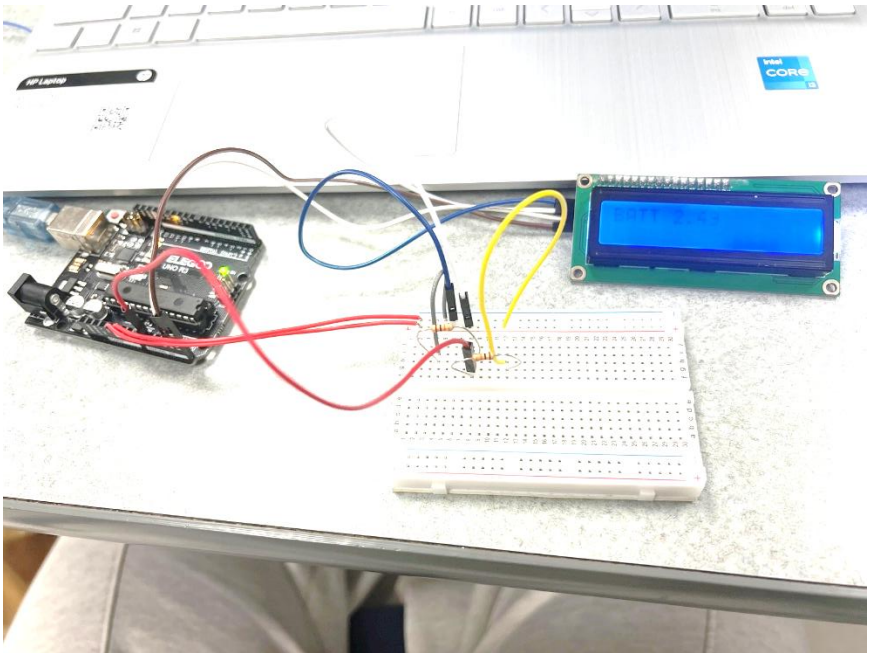
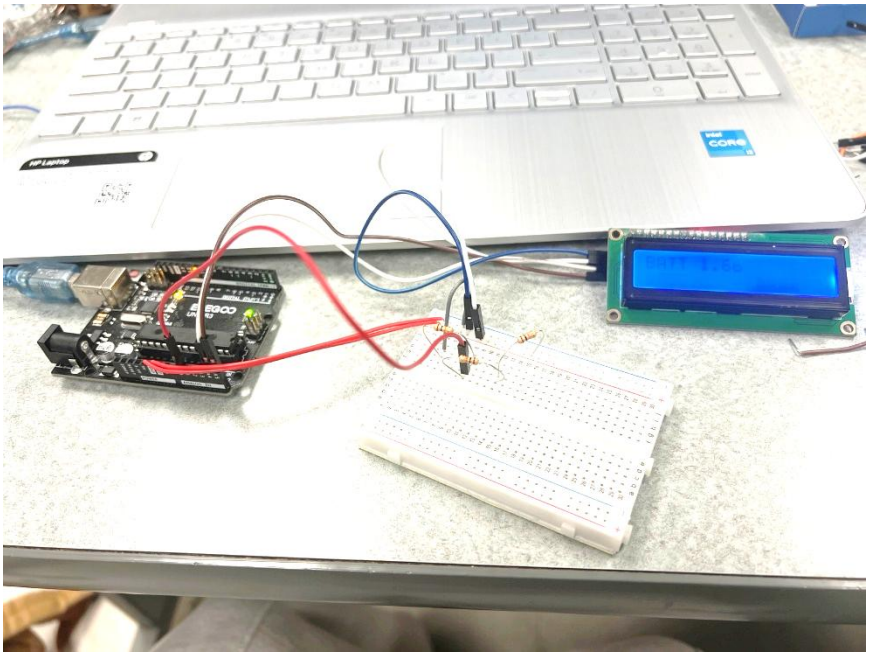


Figure 2: Arduino Voltage Measurement Circuit – Case 2



6. Conclusion

In this laboratory experiment, an Arduino-based voltage measurement system was designed using the ATmega328P microcontroller and an I2C LCD. The experiment demonstrated how it uses analog inputs to measure voltage and communicate with an external display through the I2C protocol.

Using an I2C reduced the number of required connections and simplified the circuit design. Additionally, the resistor network provided a method for controlling the voltage applied to the analog input.

Overall, the experiment provided practical experience with microcontroller interfacing, analog-to-digital conversion, circuit design, and LCD communication.